Projectiles visually update slowly when their motion is changed

MC-276317 Projectiles

Status	Resolved / Fixed
Affected	24w36a
Fixed in	24w37a
Reported	2024-09-04

STEPS TO REPRODUCE

1. Launch Minecraft 24w36a.
2. Join a world with cheats enabled.
3. Fire an arrow with a bow in any direction.
4. While the arrow is in the air, run `/data merge entity @n[type=minecraft:arrow] {Motion:[0.0d,1.0d,0.0d]}` in chat or with a command block.
5. Wait approximately one second.
6. Observe that the arrow visually continues along its old trajectory, then teleports to its true location; it may remain out of sync with its true position for another second or more before correcting again.

OBSERVED BEHAVIOUR

Actual behavior

After an airborne arrow's `Motion` NBT is changed, the entity's simulated position and motion are updated, but its rendered position and direction do not update immediately. The arrow continues to be displayed along its previous trajectory for approximately one second, often moving erratically relative to its actual position.

Once the visual state updates, the arrow may snap to its current position. In some cases, its position updates while its displayed direction or motion remains stale for another second or longer, causing it to remain visually out of sync before correcting itself again. If the arrow appears to reach the ground before the visual correction occurs, no impact sound is played because the actual entity has not reached the ground at that time.

The entity's actual position can be observed by following it with a particle command such as:

```mcfunction
execute as @e[type=minecraft:arrow] at @s run particle minecraft:electric_spark ^ ^ ^ 0 0 0 0 0 force @a
```

The particles follow the updated entity position while the arrow model continues along its old trajectory. This occurs regardless of whether `Motion` is changed with `data merge`, `data modify`, or by an in-world effect such as being hit by a wind charge. TNT explosions, however, update the arrow's displayed motion correctly.

The issue affects projectiles whose models display their flight direction, including arrows, tridents, and wither skulls. Projectiles such as snowballs, ender pearls, and wind charges do not exhibit the delayed visual update.

EXPECTED BEHAVIOUR

When an arrow, trident, wither skull, or other direction-displaying projectile has its motion changed, its visual position and direction should update immediately to match its actual Motion and follow the new trajectory. It should not continue along the previous path, teleport after a delay, or remain visually out of sync with its true location, and any resulting impact should behave and appear normally.

ENVIRONMENT

Minecraft Java Edition Version 24w36a (snapshot)

ORIGINAL DESCRIPTION

When a projectile that also displays a direction such as an arrow or a trident has its Motion changed, it will visually continue to advance in its old trajectory in a chaotic manner.

In the attached video I am using a datapack that changes arrows Motion so that they follow already landed arrows, their behavior has not changed since 24w35a. but visually they continue to follow their old trajectory for a few seconds.

Video example: https://youtu.be/-czvnLJv1do?si=1nQcCMhHf0kDCp2w, es_reproducing.mp4

(The sky is not glitched, there's just a lot of glass)

Only projectiles that display their direction seem to be affected, so arrows, tridents and wither skulls. While snowballs, ender_pearls, or wind charges are not affected.

The delay in the visual update is not tied to any command, it instead appears to be universal, it does not matter if i use data modify, data merge, or if the arrow is hit with a wind charge in mid air, the correct motion will only be displayed after one second.

TNT explosions update the arrows visual motion correctly.

Steps to reproduce

- Join any world in 24w36a with cheats enabled.

- Using a bow and an arrow, fire in any direction.

- While the arrow is in the air, use the following command in the chat or with a command block:

/data merge entity @n[type=minecraft:arrow] {Motion:[0.0d,1.0d,0.0d]}

alternatively use the command:

/data modify entity @n[type=minecraft:arrow] Motion set value [0.0d,1.0d,0.0d]

Observed behavior

The arrow flies its old trajectory for about a second, after which it teleports to its true location, sometimes the arrow teleports to its true location but does not display the correct Motion, so it remains out of sync with its true position for another second or more before teleporting again. If the arrow visually lands before correcting its position to its true location, it produces no sound when hitting the ground.

Expected behavior

The arrow immediately follows its new trajectory, displaying its position and motion correctly.

By placing a repeating command block on the ground and activating it after inserting the following command:

execute as @e[type=minecraft:arrow] at @s run particle minecr

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: https://bugs.mojang.com/browse/MC/issues/MC-276317